

Production Hardening, Observability & Cost Optimisation Sprint

Technical Design Document

Region: ap-south-2

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1. Executive Summary

The **Production Hardening, Observability & Cost Optimisation Sprint for Employee Document Vault with Role-Based Access Control** is a production-readiness engineering pass performed after the successful implementation of the serverless document management application. Rather than introducing new business functionality, this enhancement focuses on improving the operational reliability, security posture, observability, deployment automation, and cost efficiency of the existing AWS-based solution.

The enhancement integrates **Amazon CloudWatch**, **AWS X-Ray**, **Amazon SNS**, **AWS IAM**, **GitHub Actions**, **AWS Trusted Advisor**, and **Artillery** to transform the application into a production-grade cloud-native system. Comprehensive monitoring and distributed tracing have been implemented to provide real-time visibility into application performance, while CloudWatch Dashboards and Log Insights enable administrators to monitor Lambda invocations, execution duration, error rates, and overall system health.

To improve operational responsiveness, CloudWatch Alarms have been configured to detect abnormal error rates and high application latency. These alarms publish notifications through Amazon SNS, allowing administrators to receive immediate email alerts whenever operational thresholds are exceeded. A standardized incident response runbook has also been prepared to assist operators in identifying root causes and performing initial recovery actions.

Security has been strengthened by reviewing all IAM roles and replacing wildcard permissions with least-privilege access policies. Encryption at rest has been verified for Amazon S3 and Amazon DynamoDB resources, while Amazon Cognito continues to provide secure user authentication and role-based authorization. AWS Trusted Advisor has been used to validate security best practices and identify potential operational improvements.

The enhancement also introduces a complete Continuous Integration and Continuous Deployment (CI/CD) pipeline using GitHub Actions. Every code commit automatically initiates a Build → Test → Deploy workflow, ensuring that application updates are validated before deployment. Failed tests automatically block deployments, improving software quality and reducing production risk.

Performance validation has been completed through Artillery load testing, which measures application throughput, response latency, and error rates under concurrent user traffic. Cost estimation and optimization have also been performed to analyze system scalability across multiple workload scenarios and identify opportunities for reducing operational expenses.

Overall, this DevOps enhancement significantly improves the maintainability, reliability, scalability, security, and operational visibility of the Employee Document Vault, making it suitable for production deployment while following AWS Well-Architected Framework best practices.

2. Project Overview

The **Production Hardening, Observability & Cost Optimisation Sprint for Employee Document Vault with Role-Based Access Control (RBAC)** is a fully serverless AWS application that enables secure storage, retrieval, and management of employee documents. The system stores documents in Amazon S3, maintains document metadata in Amazon DynamoDB, authenticates users using Amazon Cognito, and processes all business logic through AWS Lambda functions exposed via Amazon API Gateway. Role-Based Access Control ensures that employees, managers, and HR administrators can only access documents according to their assigned permissions.

Following the successful implementation of the application's functional requirements, a comprehensive DevOps enhancement was carried out to improve production readiness without modifying the existing business logic. The enhancement focuses on operational monitoring, centralized logging, distributed tracing, automated deployment, security hardening, performance validation, and cost optimization.

Observability has been implemented using Amazon CloudWatch and AWS X-Ray, enabling developers and administrators to monitor application health, trace request execution across AWS services, and quickly diagnose failures. CloudWatch Dashboards provide real-time visualization of Lambda invocation counts, execution duration, error rates, and P95 latency, while CloudWatch Log Insights supports advanced operational log analysis.

Operational reliability has been further improved through CloudWatch Alarms integrated with Amazon SNS email notifications. These alerts enable proactive detection of application failures and high-latency conditions, reducing incident response time and improving overall service availability.

Security enhancements include a comprehensive review of IAM permissions to enforce the principle of least privilege, verification of encryption at rest for Amazon S3 and DynamoDB resources, validation of Cognito-based authentication, and assessment of AWS Trusted Advisor recommendations. Together, these measures strengthen the overall security posture of the application.

Deployment automation has been implemented through GitHub Actions, allowing source code changes pushed to the main branch to automatically trigger a Build → Test → Deploy pipeline. Automated testing ensures that only validated code is deployed to AWS Lambda, reducing manual effort and minimizing deployment errors.

Finally, Artillery-based load testing and cost analysis have been performed to evaluate application scalability, identify performance bottlenecks, estimate operational expenses under different workloads, and recommend optimization strategies for future growth.

The resulting architecture demonstrates modern DevOps engineering practices while preserving the original serverless architecture and business functionality of the Employee Document Vault system.

3. Engineering Objectives

The DevOps Enhancement focuses on transforming the existing Employee Document Vault into a production-ready cloud-native application by improving its operational reliability, security, automation, and performance. Unlike the original implementation, which primarily addressed business functionality, this engineering phase concentrates on ensuring that the application can be efficiently monitored, maintained, deployed, and scaled in real-world production environments.

The primary engineering objectives of this enhancement are as follows:

- **Improve System Observability**

Implement centralized monitoring and distributed tracing using Amazon CloudWatch and AWS X-Ray to provide complete visibility into application execution, service dependencies, request flow, and operational health.

- **Enable Proactive Monitoring**

Configure CloudWatch Dashboards and CloudWatch Alarms to continuously monitor critical operational metrics including Lambda invocation count, execution duration, error rate, and P95 response latency.

- **Strengthen Security Posture**

Review all AWS IAM roles and eliminate wildcard (*) permissions by implementing least-privilege access policies. Verify encryption at rest for Amazon S3 buckets and DynamoDB tables while maintaining secure user authentication through Amazon Cognito.

- **Improve Incident Response**

Implement automated alert notifications using Amazon SNS so that operational issues are immediately reported to administrators. Develop a standardized runbook describing likely causes, troubleshooting steps, and recovery procedures for each monitored alarm.

- **Optimize Operational Cost**

Analyze infrastructure costs under multiple workload scenarios and identify optimization opportunities such as Lambda memory tuning, lifecycle policies, log retention optimization, and efficient storage management to minimize monthly operational expenses.

- **Automate Software Deployment**

Establish a Continuous Integration and Continuous Deployment (CI/CD) pipeline using GitHub Actions to automatically build, test, and deploy AWS Lambda functions whenever changes are committed to the main source code branch.

- **Validate Production Scalability**

Perform concurrent load testing using Artillery to evaluate system throughput, latency, error rates, and resource utilization under simulated production traffic while identifying potential bottlenecks and future scalability improvements.

- **Increase Maintainability**

Provide comprehensive operational documentation including monitoring strategy, deployment procedures, security verification, load testing results, and incident response guidelines to simplify long-term system maintenance.

4. Functional Enhancements

The DevOps Enhancement introduces several operational capabilities that extend the original Employee Document Vault without modifying its business functionality. These enhancements focus on monitoring, security, deployment automation, operational visibility, and production readiness.

The enhanced system provides the following capabilities:

- Enable AWS X-Ray tracing for all AWS Lambda functions to visualize end-to-end request execution across AWS services.
- Capture distributed traces and generate AWS X-Ray Service Maps for complete request flow analysis.
- Collect detailed application logs through Amazon CloudWatch Logs.
- Execute CloudWatch Log Insights queries to analyze Lambda execution duration, operational failures, and application performance trends.
- Display real-time operational metrics through Amazon CloudWatch Dashboards.
- Monitor Lambda invocation count, execution duration, error rate, throttling events, and P95 latency.
- Generate automatic CloudWatch Alarms whenever application error rates exceed predefined thresholds.
- Trigger latency alarms when Lambda execution duration exceeds acceptable response times.
- Publish operational alerts to Amazon SNS for immediate email notification.
- Maintain documented incident response procedures for operational failures.
- Review and update IAM policies according to the Principle of Least Privilege.
- Remove wildcard (*) resource permissions from Lambda execution roles.
- Verify encryption at rest for Amazon S3 buckets using server-side encryption.

- Verify encryption at rest for Amazon DynamoDB tables using AWS-managed encryption keys.
 - Validate secure authentication and authorization through Amazon Cognito.
 - Perform AWS Trusted Advisor security and fault-tolerance assessments.
 - Estimate monthly operational costs under multiple workload scenarios.
 - Identify the highest-cost AWS service within the architecture.
 - Implement infrastructure optimization strategies to reduce operational expenses.
 - Automate software deployment using GitHub Actions.
 - Execute Build → Test → Deploy workflow for every source code update.
 - Prevent deployment whenever automated tests fail.
 - Display CI/CD pipeline status using GitHub README badges.
 - Perform concurrent load testing using Artillery.
 - Measure throughput, response latency, error rate, and application stability under sustained traffic.
 - Produce operational reports documenting monitoring results, security validation, deployment success, and performance analysis.
-

5. Non-Functional Improvements

The DevOps Enhancement significantly strengthens the quality attributes of the Employee Document Vault by improving its operational characteristics while preserving the original application functionality.

■ Scalability

The application continues to leverage fully managed AWS serverless services, allowing resources such as AWS Lambda, Amazon S3, and Amazon DynamoDB to scale automatically according to workload demands. Load testing validates system behavior under concurrent user traffic and provides recommendations for handling significantly larger workloads.

■ Reliability

CloudWatch monitoring, automated alarms, and proactive notifications improve system reliability by enabling rapid detection and resolution of operational failures. Automated deployment testing further reduces the likelihood of production errors.

■ Performance

AWS X-Ray tracing and CloudWatch performance metrics provide detailed visibility into request execution times, enabling identification of performance bottlenecks. Continuous

monitoring of P95 latency ensures that application response times remain within acceptable operational limits.

■ Security

Security has been enhanced through implementation of least-privilege IAM policies, verification of encryption at rest, secure Cognito authentication, AWS Trusted Advisor recommendations, and comprehensive security auditing. These measures reduce the attack surface while protecting sensitive employee documents and metadata.

■ Availability

Real-time monitoring and automated alerting minimize downtime by enabling administrators to respond immediately to abnormal application behavior. The serverless architecture also provides high availability through fully managed AWS services without requiring dedicated infrastructure management.

■ Observability

Comprehensive logging, distributed tracing, dashboards, alarms, and operational analytics provide complete visibility into application behavior. Administrators can monitor system health, diagnose failures, and analyze performance trends from a centralized monitoring platform.

■ Maintainability

Centralized logging, documented operational procedures, automated deployment pipelines, and structured monitoring simplify application maintenance and reduce the operational effort required to support production workloads.

■ Cost Efficiency

Operational costs are continuously evaluated through workload-based cost estimation and optimization techniques including Lambda resource tuning, storage lifecycle policies, and efficient monitoring configuration. These improvements ensure that the application maintains a cost-effective serverless architecture while supporting increasing workloads.

■ Deployment Automation

The GitHub Actions CI/CD pipeline automates software delivery by executing build, testing, and deployment stages for every code update. Automated validation reduces manual deployment effort, improves consistency, and increases release reliability.

6. Enhanced High-Level Architecture

The DevOps Enhancement preserves the existing serverless architecture of the Employee Document Vault while integrating additional AWS services that improve observability, operational monitoring, security, deployment automation, and performance validation. These enhancements operate alongside the existing application without modifying its business logic.

The enhanced architecture follows the workflow below:

1. The user authenticates through **Amazon Cognito User Pool** and receives a JSON Web Token (JWT) for secure access.
2. The authenticated user accesses the web application hosted in an Amazon S3 static website bucket.
3. User requests are routed through **Amazon API Gateway**, which validates authentication before forwarding requests to the appropriate AWS Lambda function.
4. AWS Lambda functions perform business operations such as document upload, download, listing, and deletion while interacting with Amazon S3 and Amazon DynamoDB.
5. Every Lambda invocation generates execution logs that are automatically stored in **Amazon CloudWatch Logs**.
6. AWS X-Ray captures distributed traces for each Lambda invocation, enabling end-to-end request visualization through the X-Ray Service Map.
7. Amazon CloudWatch continuously collects operational metrics, including invocation count, execution duration, error rate, concurrent executions, and P95 latency.
8. CloudWatch Dashboards visualize these metrics in real time, providing centralized monitoring for administrators.
9. CloudWatch Alarms continuously evaluate operational thresholds. If the Lambda error rate exceeds 5% or P95 latency exceeds three seconds, an alarm is triggered automatically.
10. Alarm notifications are published to an **Amazon SNS Topic**, which immediately sends email notifications to system administrators.
11. Source code changes pushed to the GitHub repository automatically initiate the **GitHub Actions CI/CD pipeline**.
12. The pipeline executes Build → Test → Deploy stages. Successful builds automatically deploy updated Lambda functions using the AWS CLI, while failed tests prevent deployment.
13. AWS Trusted Advisor periodically evaluates the AWS environment to identify security risks, service limits, and operational best-practice recommendations.
14. Artillery executes performance tests against the primary API endpoint to measure throughput, latency, and application stability under concurrent workloads.

The enhanced architecture enables continuous monitoring, automated deployment, proactive alerting, and production-ready operational management while maintaining the original serverless application design.

7. Enhanced Component Design

7.1 Amazon CloudWatch

Amazon CloudWatch acts as the centralized monitoring platform for the enhanced application.

Responsibilities

- Collect Lambda execution logs.
- Monitor invocation count.
- Measure execution duration.
- Track application error rates.
- Monitor concurrent Lambda executions.
- Record P95 response latency.
- Store operational metrics.
- Generate dashboards.
- Trigger CloudWatch Alarms.

CloudWatch enables administrators to monitor application health in real time and rapidly identify operational anomalies.

7.2 AWS X-Ray

AWS X-Ray provides distributed tracing across the serverless application.

Responsibilities

- Trace every Lambda invocation.
- Measure service execution time.
- Identify bottlenecks.
- Generate Service Maps.
- Visualize request flow.
- Assist performance troubleshooting.

The X-Ray Service Map illustrates how requests travel between API Gateway, Lambda, DynamoDB, and Amazon S3, allowing developers to isolate slow or failing components.

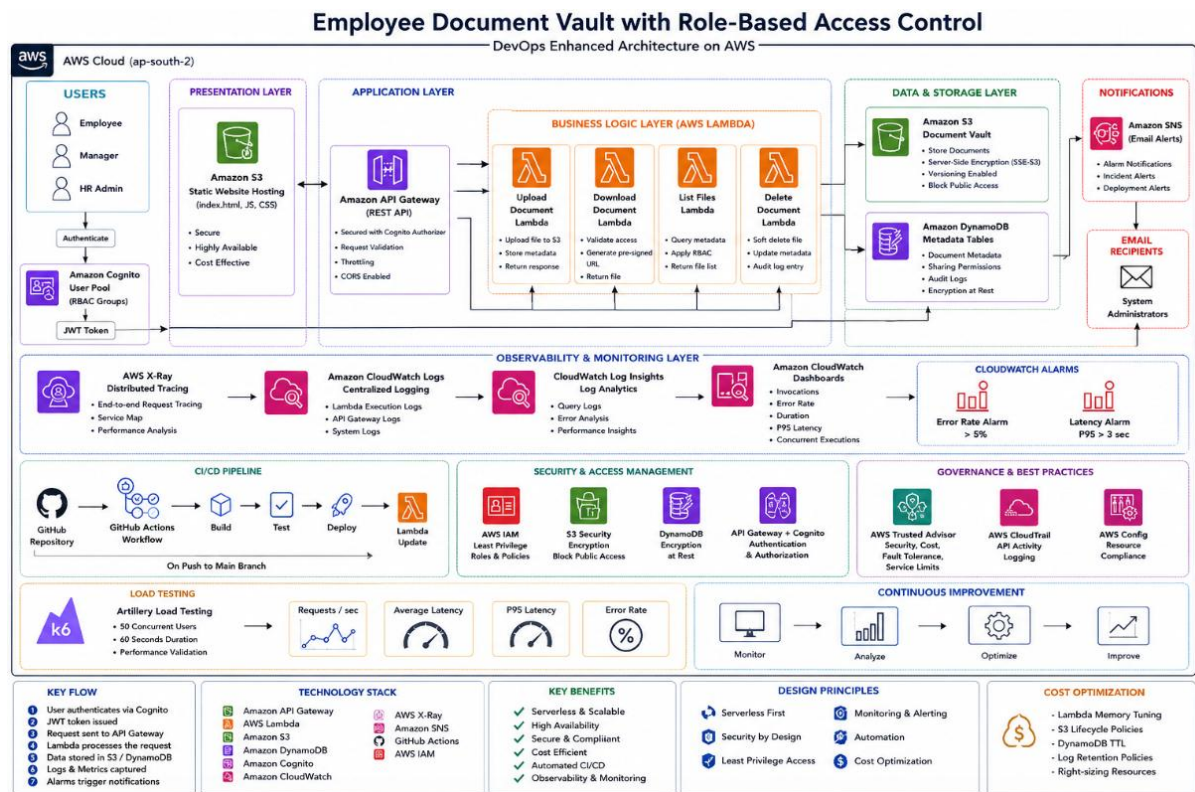


Figure 1: AWS Architecture of DevOps Enhancements of Employee Document Vault

7.3 Amazon SNS

Amazon Simple Notification Service (SNS) provides automated operational alerting.

Responsibilities

- Receive CloudWatch Alarm notifications.
- Publish email alerts.
- Notify administrators immediately.
- Support incident response.

Whenever operational thresholds are exceeded, CloudWatch automatically publishes alarm messages to the SNS topic, ensuring rapid notification without manual intervention.

7.4 AWS IAM

AWS Identity and Access Management secures all AWS resources by implementing least-privilege permissions.

Responsibilities

- Manage Lambda execution roles.

- Restrict S3 bucket permissions.
- Restrict DynamoDB permissions.
- Remove wildcard (*) access.
- Control CloudWatch permissions.
- Control X-Ray tracing permissions.

Every Lambda function receives only the minimum permissions necessary to perform its assigned task, significantly reducing the application's security exposure.

7.5 Amazon Cognito

Amazon Cognito continues to provide authentication and authorization for all application users.

Responsibilities

- Authenticate users.
- Issue JWT tokens.
- Maintain user groups.
- Enforce Role-Based Access Control.
- Protect API Gateway endpoints.

Only authenticated users with valid JWT tokens can invoke backend APIs.

7.6 GitHub Actions

GitHub Actions automates the software delivery lifecycle.

Responsibilities

- Detect source code changes.
- Execute Build stage.
- Run automated tests.
- Deploy Lambda functions.
- Block deployment on test failure.
- Maintain deployment consistency.

The automated pipeline eliminates manual deployment activities while improving software quality and release reliability.

7.7 AWS Trusted Advisor

AWS Trusted Advisor continuously evaluates AWS resource configuration according to AWS best practices.

Responsibilities

- Review security configuration.
- Check IAM usage.
- Verify S3 permissions.
- Monitor service limits.
- Recommend operational improvements.

Trusted Advisor assists administrators in maintaining a secure and reliable production environment.

7.8 Artillery

Artillery is used to validate application performance under concurrent user workloads.

Responsibilities

- Generate concurrent API requests.
- Measure throughput.
- Measure response latency.
- Identify throttling.
- Detect capacity bottlenecks.
- Produce performance reports.

Load testing confirms that the application remains stable under increased traffic while identifying future scalability improvements.

8. Observability Design

Observability is one of the most significant improvements introduced during the DevOps Enhancement. The application has been instrumented using AWS X-Ray and Amazon CloudWatch to provide complete visibility into system behaviour, request execution, and operational performance.

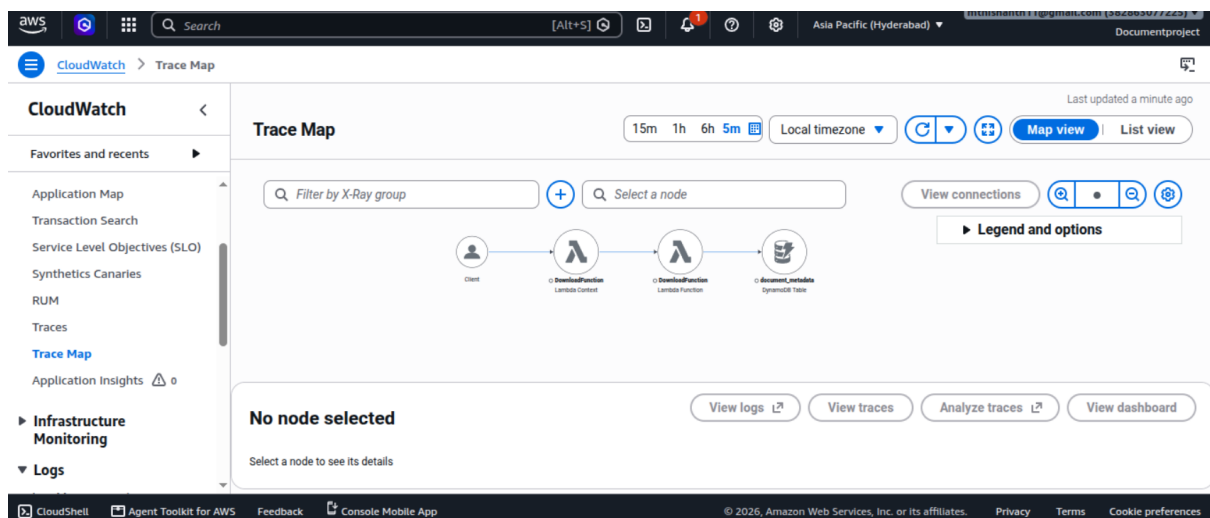
Distributed Tracing

AWS X-Ray tracing has been enabled for all AWS Lambda functions. Every incoming request generates a trace that records execution time, downstream service calls, exceptions, and processing delays.

The generated Service Map visually represents communication between:

- Amazon API Gateway
- AWS Lambda
- Amazon DynamoDB
- Amazon S3
- Amazon Cognito

This visualization enables developers to quickly identify latency issues, failed requests, and service dependencies.



Centralized Logging

All Lambda execution logs are automatically stored in Amazon CloudWatch Logs.

Each log entry records:

- Request ID
- Timestamp
- Lambda Function Name
- Execution Duration
- Memory Usage
- Success Status
- Error Messages

- Exception Details

Centralized logging simplifies debugging and operational analysis.

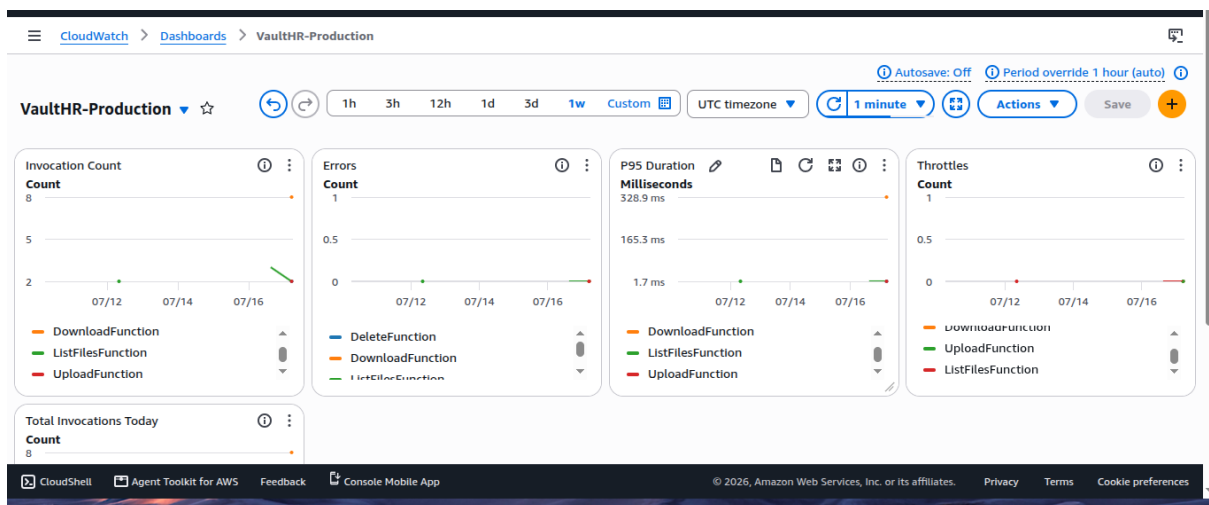
CloudWatch Log Insights

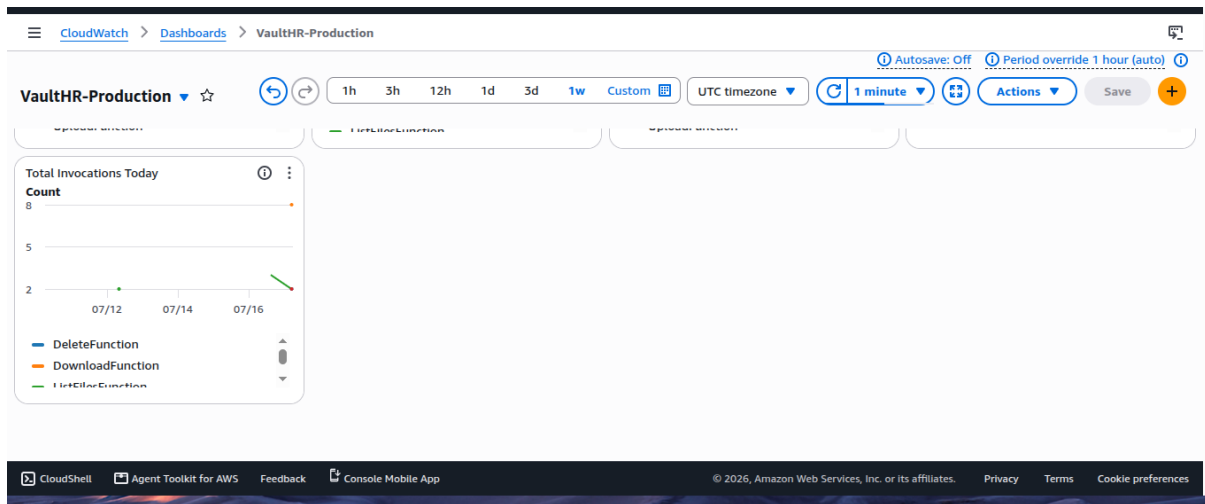
CloudWatch Log Insights enables administrators to execute advanced queries across Lambda log groups.

Typical operational queries include:

- Average Lambda execution duration
- Total failures during the current day
- Most frequently invoked Lambda functions
- Timeout analysis
- Exception frequency
- Error distribution by function

These queries significantly reduce troubleshooting time by enabling rapid analysis of operational data.





Operational Metrics

CloudWatch continuously collects operational metrics including:

Metric	Purpose
Invocation Count	Total Lambda executions
Error Rate	Percentage of failed invocations
Duration	Lambda execution time
P95 Latency	Performance monitoring
Throttles	Detect concurrency limitations
Concurrent Executions	Monitor workload scaling

These metrics provide administrators with a comprehensive understanding of application behaviour.

Benefits of the Observability Layer

The implemented observability solution provides several operational advantages:

- Complete visibility into request execution.
- Faster troubleshooting of production issues.
- Improved application reliability.
- Early detection of performance degradation.
- Real-time operational monitoring.
- Better capacity planning through historical metrics.

- Reduced Mean Time to Detect (MTTD) and Mean Time to Recover (MTTR).
- Support for proactive system maintenance.

9. Logging and Monitoring Strategy

Amazon CloudWatch serves as the centralized monitoring and logging platform for the Employee Document Vault after the DevOps enhancement. All AWS Lambda functions continuously send execution logs and performance metrics to CloudWatch, allowing administrators to monitor application health, diagnose failures, and analyse system behaviour from a single location.

The logging strategy follows a structured approach where every Lambda invocation generates detailed operational information. Logs include request identifiers, execution timestamps, execution duration, memory utilization, API request details, and exception messages. These records enable developers to trace the lifecycle of each request and quickly identify operational issues.

The monitored Lambda functions include:

- Upload Lambda
- Download Lambda
- List Files Lambda
- Delete Lambda

Each function records important execution events including:

Upload Lambda

- Lambda invocation started
- User authentication successful
- JWT validation completed
- Document metadata validated
- File uploaded successfully
- Metadata stored in DynamoDB
- Upload completed
- Exception details (if any)

Download Lambda

- Download request received
- User authorization verified

- Document metadata retrieved
 - Pre-signed URL generated
 - Download completed
 - Permission denied events
 - Execution exceptions
-

List Files Lambda

- User authenticated
 - RBAC validation successful
 - Metadata query executed
 - Files returned
 - Query execution duration
 - Error messages
-

Delete Lambda

- Delete request received
 - Authorization validated
 - Soft delete initiated
 - DynamoDB metadata updated
 - Audit log generated
 - Delete completed successfully
-

CloudWatch Log Insights

CloudWatch Log Insights enables administrators to execute advanced analytical queries against Lambda log groups without exporting log data.

The following operational queries have been configured:

Query 1 – Average Lambda Execution Duration

fields @timestamp, @duration

| stats avg(@duration) as AverageDuration by @log

Purpose

- Monitor average execution time
- Identify slow Lambda functions
- Detect performance degradation

Query 2 – Total Errors Generated Today

fields @timestamp,@message

| filter @message like /ERROR/

| stats count() as TotalErrors

Purpose

- Count daily failures
- Measure application stability
- Support incident investigation

Query 3 – Lambda Invocations

fields @timestamp

| stats count() by @log

Purpose

- Measure workload
- Compare Lambda usage
- Identify heavily used APIs

The screenshot displays the AWS CloudWatch Logs console interface. At the top, there's a navigation bar with 'Query 1', 'Help', 'Feedback', and 'Preferences'. Below this is a search bar with the placeholder 'Ask AI to write a query...'. To the right of the search bar are filters for time range (5m, 30m, 1h, 3h, 12h) and a 'Last 1 week' button. Below the search bar are tabs for 'Log group tags', 'Data sources', and 'Facets'. The 'Data sources' tab is active, showing filters for '@aws.region', '@data_format', '@data_source_name', and '@data_source_type'. The main area is titled 'Query by Log group' and contains a KQL query editor. The query is as follows:

```
1 > SOURCE "arn:aws:logs:ap-south-2:382863077225:log-group:/aws/lambda/DeleteFunction" START=-664860s END=0s | ...
6 filter @type = "REPORT"
7 | stats avg(@duration) as avg_ms,
8 | .....max(@duration) as max_ms,
9 | .....min(@duration) as min_ms,
10 | .....count() as invocations
11 | by @log
12 | sort avg_ms desc
```

Below the query editor, there are buttons for 'Visualization', 'Analyze patterns', and 'Run' (with a 'Cancel' link). A status bar at the bottom indicates 'Complete: Showing 1 of 2 matched. Query executed for 4 log groups.' and '7 records (1.1 KB) scanned in 1.4s @ 5 records/s (783 B/s)'. The bottom of the screen shows a timeline view with a date 'Jul 11, 2026, 09:30:00 AM'.

Centralized Monitoring Benefits

The CloudWatch logging strategy provides:

- Centralized operational visibility
 - Faster debugging
 - Reduced Mean Time To Detect (MTTD)
 - Improved troubleshooting efficiency
 - Historical log retention
 - Operational analytics
 - Performance monitoring
 - Failure investigation
-

10. CloudWatch Dashboard Design

A centralized Amazon CloudWatch Dashboard has been created to provide real-time operational visibility for the Employee Document Vault. The dashboard aggregates key Lambda metrics into a single monitoring interface, allowing administrators to assess application health without inspecting individual resources.

The dashboard continuously refreshes operational metrics collected from AWS Lambda and CloudWatch.

Dashboard Widgets

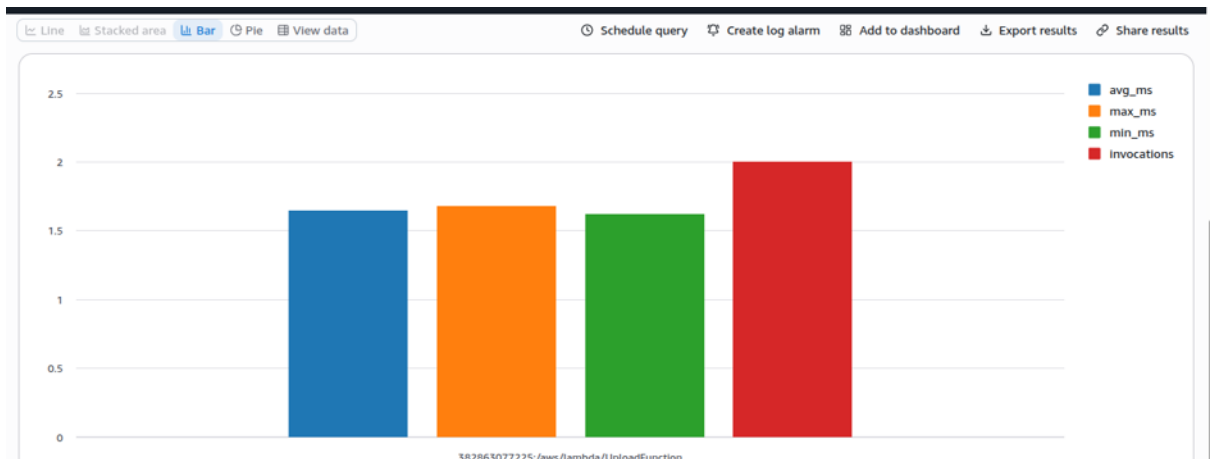
1. Lambda Invocation Count

Purpose

Displays the total number of Lambda executions over time.

Benefits

- Monitor application usage
- Detect sudden traffic spikes
- Verify successful API activity



2. Error Count

Purpose

Displays total Lambda execution failures.

Benefits

- Detect runtime exceptions
- Measure application stability
- Trigger incident investigation

3. Error Rate

Purpose

Calculates the percentage of failed invocations.

Threshold

Greater than **5%**

Benefits

- Detect abnormal behaviour
 - Trigger CloudWatch Alarm
 - Support SLA monitoring
-

4. Average Duration

Purpose

Displays average Lambda execution time.

Benefits

- Identify slow functions
- Compare performance trends
- Detect increasing execution time

5. P95 Duration

Purpose

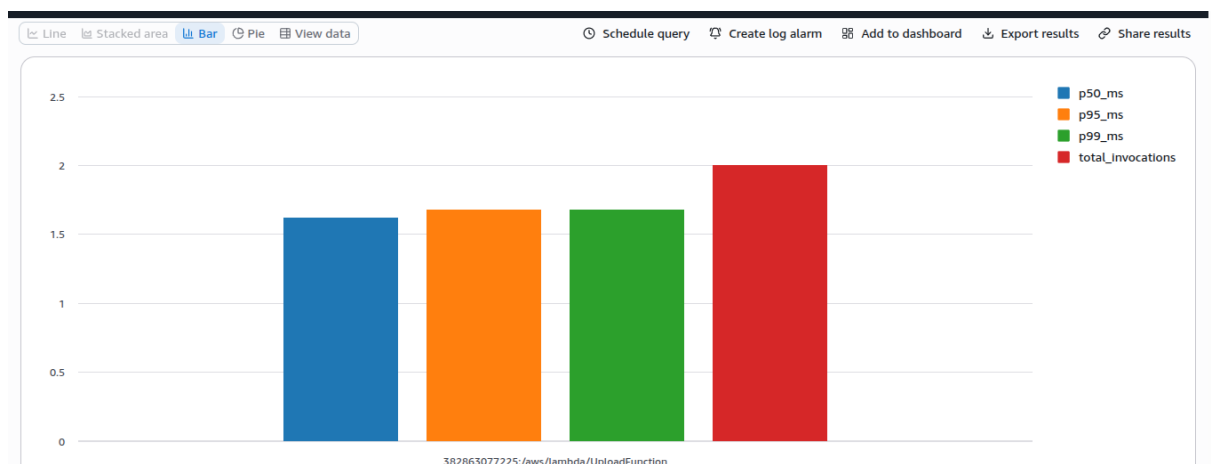
Displays the 95th percentile execution latency.

Threshold

Greater than **3 seconds**

Benefits

- Detect high latency
- Measure user experience
- Trigger latency alarm



6. Concurrent Executions

Purpose

Displays simultaneous Lambda executions.

Benefits

- Monitor scaling behaviour
 - Detect concurrency spikes
 - Identify potential throttling
-

7. Throttles

Purpose

Displays throttled Lambda requests.

Benefits

- Identify concurrency limitations
 - Evaluate scalability
 - Recommend resource optimization
-

Dashboard Advantages

The CloudWatch Dashboard provides:

- Single-pane operational visibility
- Real-time monitoring
- Performance analysis
- Capacity planning
- Failure detection
- Historical trend analysis
- Improved operational awareness

Figure X: *CloudWatch Dashboard showing Invocation Count, Error Rate, Average Duration, P95 Latency, and Concurrent Executions.*

(Insert your CloudWatch Dashboard screenshot here.)

11. Alerting and Incident Response

To ensure proactive monitoring of production workloads, CloudWatch Alarms have been configured to automatically detect operational anomalies and notify administrators through

Amazon SNS. These alerts reduce response time and improve service reliability by enabling immediate action when performance thresholds are exceeded.

11.1 CloudWatch Alarm 1 – Error Rate Alarm

Alarm Name: Lambda-Error-Rate

Condition

Error Rate > **5%** of total Lambda invocations.

Monitoring Metric

AWS/Lambda → Errors

Evaluation Period

5 minutes

Alarm Action

Publish notification to Amazon SNS Topic.

Notification Method

Email

Purpose

- Detect application failures
- Identify runtime exceptions
- Prevent prolonged service degradation

11.2 CloudWatch Alarm 2 – High Latency Alarm

Alarm Name: Lambda-P95-Latency

Condition

P95 Duration > **3 seconds**

Monitoring Metric

AWS/Lambda → Duration (P95)

Evaluation Period

5 minutes

Alarm Action

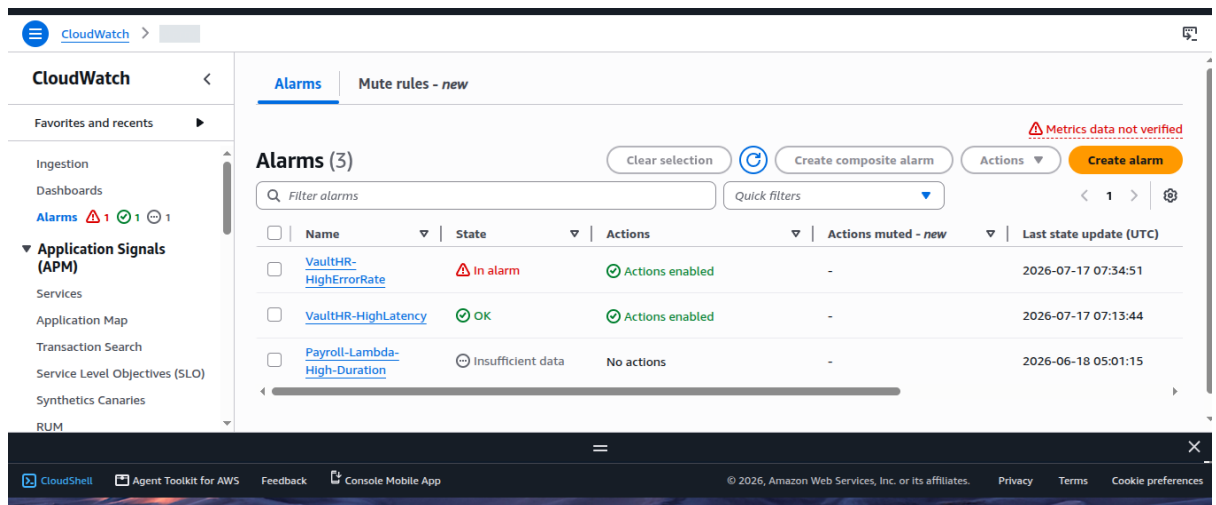
Publish notification to Amazon SNS Topic.

Notification Method

Email

Purpose

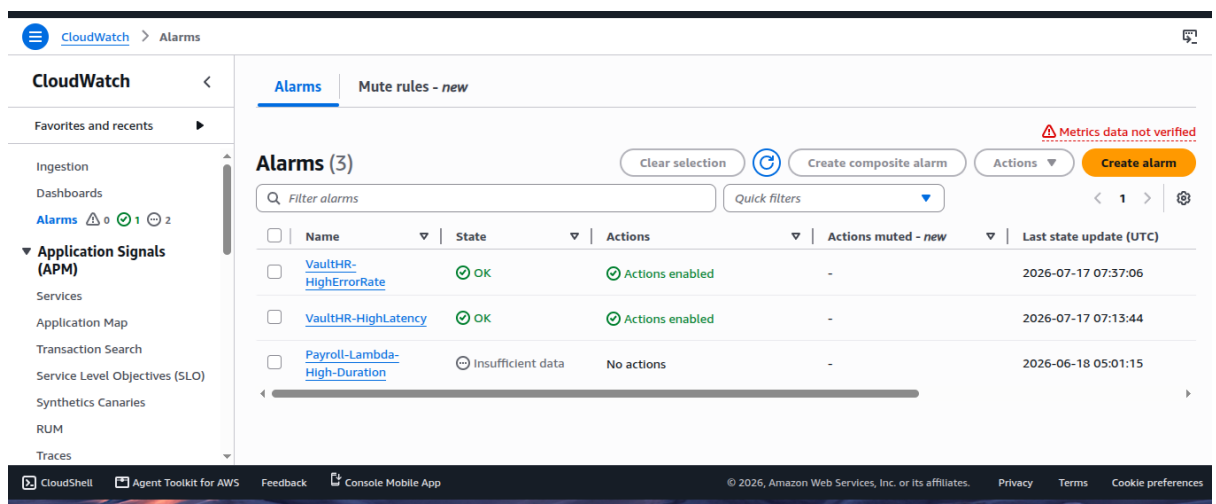
- Detect slow application response
- Identify performance bottlenecks
- Maintain acceptable user experience



The screenshot shows the AWS CloudWatch Alarms console. The left sidebar contains navigation links for CloudWatch, Favorites and recents, Ingestion, Dashboards, Alarms (1), Application Signals (APM), Services, Application Map, Transaction Search, Service Level Objectives (SLO), Synthetics Canaries, and RUM. The main content area is titled 'Alarms (3)' and includes a search bar, a 'Filter alarms' dropdown, and a 'Quick filters' dropdown. A table lists the alarms:

Name	State	Actions	Actions muted - new	Last state update (UTC)
VaultHR-HighErrorRate	In alarm	Actions enabled	-	2026-07-17 07:34:51
VaultHR-HighLatency	OK	Actions enabled	-	2026-07-17 07:13:44
Payroll-Lambda-High-Duration	Insufficient data	No actions	-	2026-06-18 05:01:15

The bottom of the console shows the footer with links to CloudShell, Agent Toolkit for AWS, Feedback, and Console Mobile App, along with copyright information for Amazon Web Services, Inc.



This screenshot is similar to the one above, showing the AWS CloudWatch Alarms console. The left sidebar is the same, but the 'Alarms' link now shows '0' instead of '1'. The main content area shows the same three alarms, but their states have changed:

Name	State	Actions	Actions muted - new	Last state update (UTC)
VaultHR-HighErrorRate	OK	Actions enabled	-	2026-07-17 07:37:06
VaultHR-HighLatency	OK	Actions enabled	-	2026-07-17 07:13:44
Payroll-Lambda-High-Duration	Insufficient data	No actions	-	2026-06-18 05:01:15

The footer remains the same, showing links to CloudShell, Agent Toolkit for AWS, Feedback, and Console Mobile App, along with copyright information for Amazon Web Services, Inc.

Amazon SNS Notification Flow

The notification workflow is illustrated below:

CloudWatch Metrics



CloudWatch Alarm

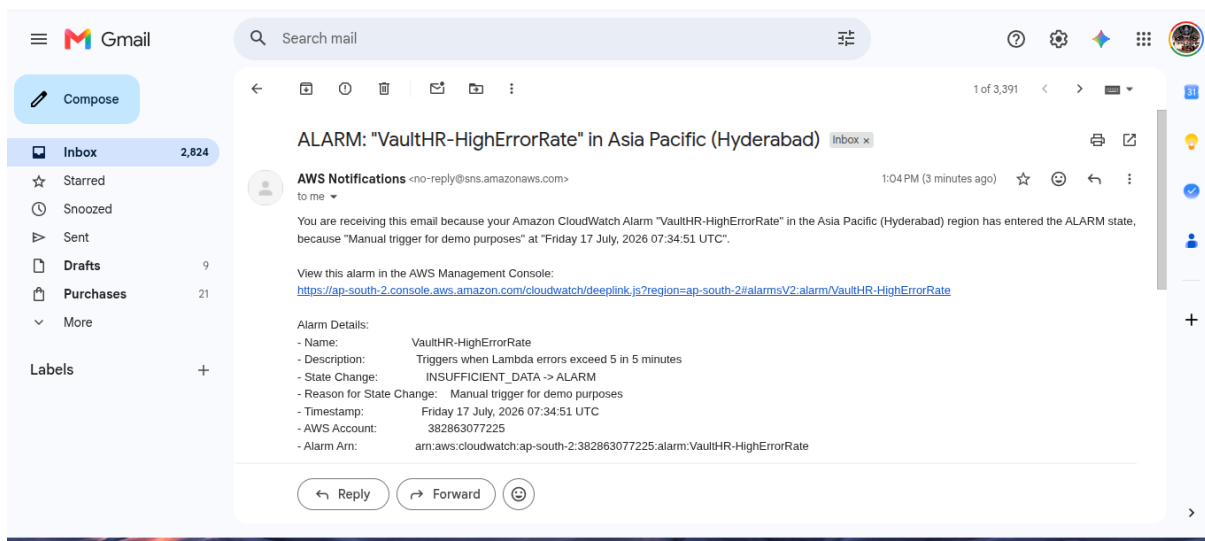


Amazon SNS Topic



Administrator Email

Whenever an alarm enters the **ALARM** state, CloudWatch automatically publishes a notification to the configured SNS topic. The SNS topic then delivers an email alert to the registered administrator, enabling rapid awareness of operational issues.



Incident Response Runbook

The following runbook provides first-response guidance for common operational alerts.

Alarm	Meaning	Possible Causes	Initial Response
Lambda Error	More than 5% of Lambda	Invalid input, IAM permission issues,	Review CloudWatch Logs, inspect Lambda error messages,

Rate > 5%	invocations are failing.	DynamoDB errors, application bugs, timeout exceptions.	verify IAM policies, confirm DynamoDB and S3 availability, and redeploy the application if necessary.
P95 Latency > 3 Seconds	Application response time exceeds the acceptable threshold.	High traffic, insufficient Lambda memory, slow DynamoDB operations, network delays, cold starts.	Check Lambda duration metrics, review X-Ray traces, verify DynamoDB performance, optimize Lambda memory settings, and monitor subsequent executions.

Incident Recovery Procedure

1. Receive the SNS email notification.
2. Open the CloudWatch Alarm dashboard.
3. Identify the affected Lambda function.
4. Review CloudWatch Logs for detailed error messages.
5. Analyse AWS X-Ray traces to locate latency or failures.
6. Verify IAM permissions and AWS service health.
7. Apply the necessary corrective action.
8. Confirm that the alarm returns to the **OK** state.
9. Document the incident and corrective measures for future reference.

Excellent! Now we'll cover the remaining technical sections that correspond to the **Security Hardening Audit, Cost Analysis, and CI/CD Pipeline** deliverables.

12. Security Hardening Design

Security hardening was performed to strengthen the Employee Document Vault by applying AWS security best practices and the principle of least privilege. The objective was to reduce the application's attack surface, protect sensitive employee documents, and ensure compliance with production security standards without changing the existing business functionality.

The security enhancement focuses on Identity and Access Management (IAM), encryption at rest, authentication, authorization, and continuous security assessment.

12.1 IAM Least-Privilege Implementation

Initially, several Lambda execution roles contained wildcard (*) permissions that granted unrestricted access to AWS resources. As part of the security audit, these permissions were reviewed and replaced with resource-specific policies.

Before Hardening

Characteristics:

- Wildcard (*) resource permissions
- Broad access to Amazon S3
- Full DynamoDB access
- Excessive CloudWatch permissions
- Increased security risk

Example Policy

```
{
  "Effect": "Allow",
  "Action": [
    "s3:*",
    "dynamodb:*"
  ],
  "Resource": "*"
}
```

The screenshot displays the AWS IAM console interface for the role **UploadFunction-role-012ib102**. The **Permissions** tab is active, showing a list of four attached policies. The role was created on June 27, 2026, and has a maximum session duration of 1 hour. The policies listed are:

Policy name	Type	Attached entities
AWSLambdaBasicExecutionRole-8f4e0411-a521-441b-907f-a...	Customer managed	1
DeleteFunctionPolicy	Customer inline	0
DownloadFunctionPolicy	Customer inline	0
ListFilesFunctionPolicy	Customer inline	0

Step 1
Modify permissions in DownloadFunctionPolicy

Step 2
Review and save

Modify permissions in DownloadFunctionPolicy

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "s3:GetObject",
7       "Resource": "arn:aws:s3:::employee-document-vault/*"
8     },
9     {
10      "Effect": "Allow",
11      "Action": [
12        "dynamodb:GetItem",
13        "dynamodb:Scan"
14      ],
15      "Resource": [
16        "arn:aws:dynamodb:ap-south-2:*:table/document_metadata",
17        "arn:aws:dynamodb:ap-south-2:*:table/manager_mapping"
18      ]
19    },
20    {
21      "Effect": "Allow",
22      "Action": "dynamodb:PutItem",
23      "Resource": "arn:aws:dynamodb:ap-south-2:*:table/audit_log"
24    }
25  ]
26 }
```

Edit statement Remove

Add actions

Choose a service

Search services

Included

DynamoDB

Available

AI Operations

AMP

API Gateway

API Gateway V2

ARC Region switch

ARC Zonal Shift

ASC

Access Analyzer

Step 1
Modify permissions in DeleteFunctionPolicy

Step 2
Review and save

Modify permissions in DeleteFunctionPolicy

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7        "dynamodb:GetItem",
8        "dynamodb:UpdateItem"
9      ],
10      "Resource": "arn:aws:dynamodb:ap-south-2:*:table/document_metadata"
11    },
12    {
13      "Effect": "Allow",
14      "Action": "dynamodb:PutItem",
15      "Resource": "arn:aws:dynamodb:ap-south-2:*:table/audit_log"
16    }
17  ]
18 }
```

Edit statement Remove

Add actions

Choose a service

Search services

Included

DynamoDB

Available

AI Operations

AMP

API Gateway

API Gateway V2

ARC Region switch

ARC Zonal Shift

ASC

Access Analyzer

Add a resource

Add

Step 1
Modify permissions in ListFilesFunctionPolicy

Step 2
Review and save

Modify permissions in ListFilesFunctionPolicy

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7        "dynamodb:Scan",
8        "dynamodb:Query"
9      ],
10      "Resource": [
11        "arn:aws:dynamodb:ap-south-2:*:table/document_metadata",
12        "arn:aws:dynamodb:ap-south-2:*:table/manager_mapping",
13        "arn:aws:dynamodb:ap-south-2:*:table/audit_log"
14      ]
15    },
16    {
17      "Effect": "Allow",
18      "Action": "dynamodb:PutItem",
19      "Resource": "arn:aws:dynamodb:ap-south-2:*:table/audit_log"
20    }
21  ]
22 }
```

Edit statement Remove

Add actions

Choose a service

Search services

Included

DynamoDB

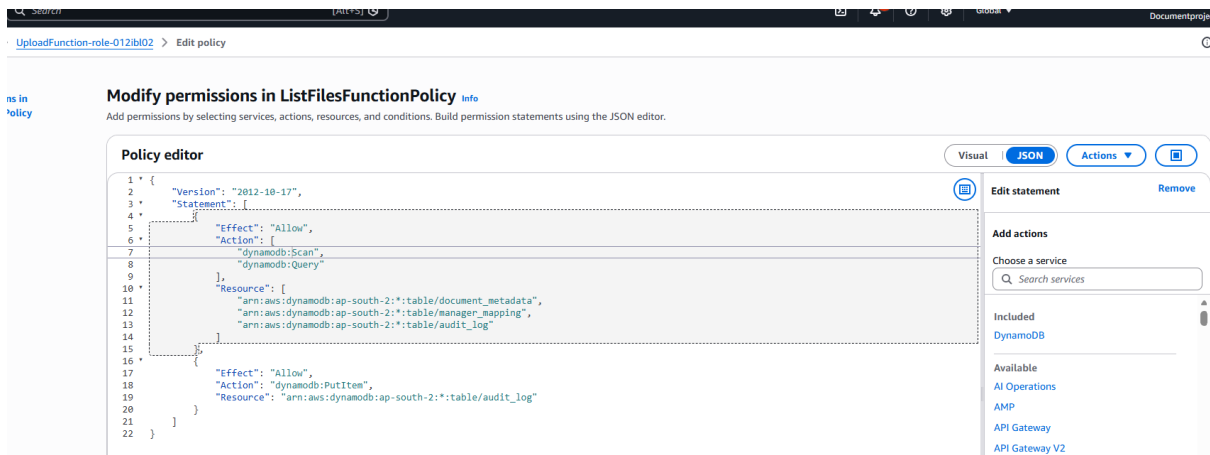
Available

AI Operations

AMP

API Gateway

API Gateway V2



After Hardening

Characteristics:

- Resource-level permissions
- Bucket-specific S3 access
- Table-specific DynamoDB access
- Restricted CloudWatch permissions
- Principle of Least Privilege

Example Policy

```
{
  "Effect": "Allow",
  "Action": [
    "s3:GetObject",
    "s3:PutObject"
  ],
  "Resource": [
    "arn:aws:s3:::vaulthur/*"
  ]
}
```

The updated IAM configuration significantly reduces unauthorized access while ensuring that each Lambda function has only the permissions required to perform its designated responsibilities.

12.2 Amazon S3 Encryption

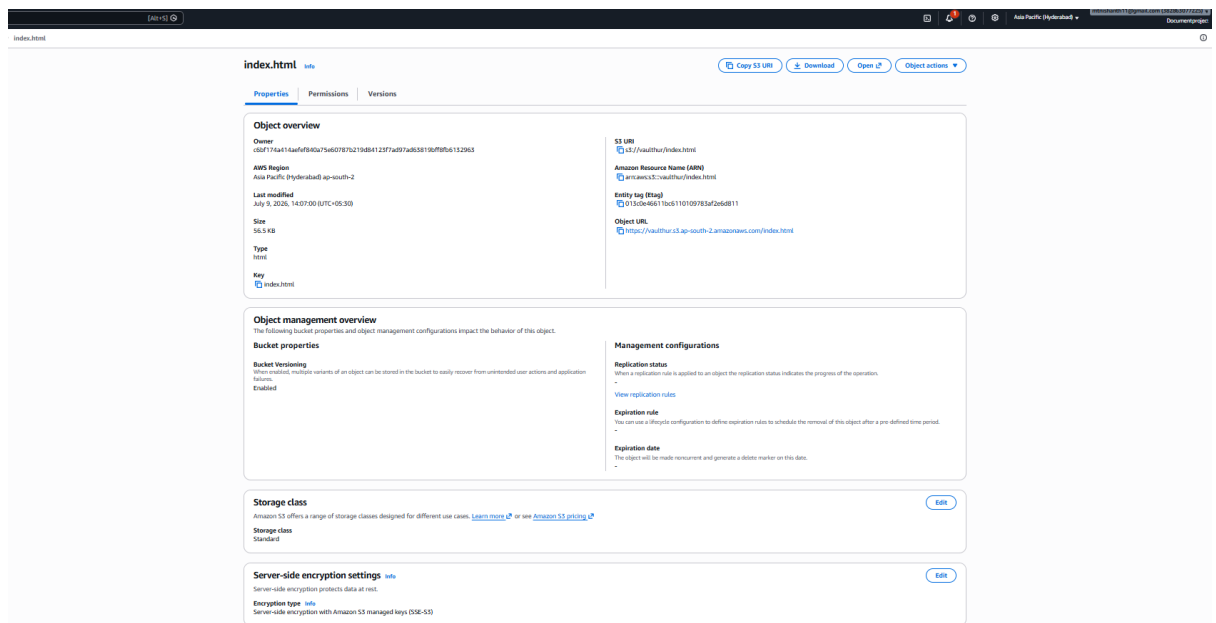
Amazon S3 server-side encryption has been enabled to protect all stored employee documents.

Configuration:

- Encryption at Rest: **Enabled**
- Encryption Type: **SSE-S3 (AWS Managed Keys)**
- Public Access Block: Enabled
- Bucket Versioning: Enabled

Benefits:

- Protects stored documents
- Meets AWS security best practices
- No application code changes required



12.3 Amazon DynamoDB Encryption

All DynamoDB tables are encrypted using AWS-managed encryption keys.

Tables:

- document_metadata
- audit_log
- manager_mapping

Configuration:

- Encryption at Rest: Enabled
- Key Management: AWS Owned Key

Benefits:

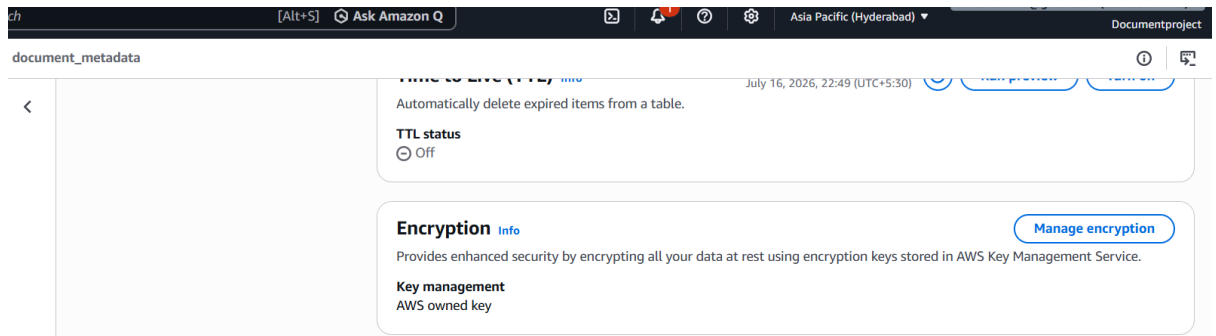
- Transparent encryption
- No performance impact
- Automatic key management

The screenshot shows the AWS IAM console interface for the 'manager_mapping' table. The table is active and has 12,000 items. The configuration panel on the right includes:

- Deletion protection**: Protected the table from being deleted unintentionally. When this setting is on, you can't delete the table. Status: Off. [Turn on](#)
- Time to Live (TTL)**: Automatically delete expired items from a table. Last updated: July 16, 2026, 22:44 (UTC+5:30). Status: Off. [Run preview](#) [Turn on](#)
- Encryption**: Provides enhanced security by encrypting all your data at rest using encryption keys stored in AWS Key Management Service. [Manage encryption](#)
- Key management**: AWS owned key

The screenshot shows the AWS IAM console interface for the 'audit_log' table. The table is active and has 12,000 items. The configuration panel on the right includes:

- Deletion protection**: Protected the table from being deleted unintentionally. When this setting is on, you can't delete the table. Status: Off. [Turn on](#)
- Time to Live (TTL)**: Automatically delete expired items from a table. Last updated: July 16, 2026, 22:48 (UTC+5:30). Status: Off. [Run preview](#) [Turn on](#)
- Encryption**: Provides enhanced security by encrypting all your data at rest using encryption keys stored in AWS Key Management Service. [Manage encryption](#)
- Key management**: AWS owned key
- Tags (0)**: Tags are labels you assign to AWS resources that allow you to manage, identify, organize, search for, and filter those resources. [Manage tags](#)



12.4 Amazon Cognito Authentication

Amazon Cognito secures user authentication and authorization.

Responsibilities:

- User authentication
- JWT token generation
- User group management
- Role-Based Access Control
- Secure API access

Only authenticated users possessing valid JWT tokens are permitted to invoke backend APIs through Amazon API Gateway.

12.5 AWS Trusted Advisor Assessment

AWS Trusted Advisor was used to evaluate the AWS environment against recommended operational best practices.

The assessment included:

- Security checks
- Fault tolerance
- Service limits
- Cost optimization
- Performance recommendations

Any identified security recommendations were reviewed and resolved before final deployment.

Trusted Advisor

Security

Recommendations

Cost optimization

Performance

Security

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Service limits

☐	ap-south-1	ap-south-1	sas-images-bucket-sa2863077225-ap-south-1-an	No	No	✔	No bucket policy
☐	ap-south-1	ap-south-1	demoarch-382863077225-ap-south-1-an	No	No	✔	No bucket policy
☐	ap-south-1	ap-south-1	sas-reports-bucket-382863077225-ap-south-1-an	No	No	✔	No bucket policy
☐	ap-south-1	ap-south-1	sas-reports-bucket1-382863077225-ap-south-1-an	No	No	✔	No bucket policy
☐	ap-south-2	ap-south-2	payroll-payslips-382863077225-ap-south-2-an	No	No	✔	No bucket policy
☐	ap-south-2	ap-south-2	employee-document-vault-team2	No	No	✔	No bucket policy

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Amazon RDS Public Snapshots

Checks the permission settings for your Amazon Relational Database Service (Amazon RDS) DB snapshots and alerts you if any snapshots are marked as public.

0 RDS snapshots are marked as public.

Last updated: 14 minutes ago

MFA on Root Account

Checks the root user credentials of an account and warns if multi-factor authentication (MFA) is not enabled. For increased security, we recommend that you protect your account by using MFA, which requires a user to enter a unique authentication code from their MFA hardware or virtual device when interacting with the AWS Management Console and associated websites.

Note

For your AWS Organizations management account, AWS requires multi-factor authentication (MFA) for the root user when accessing the AWS Management Console.

For your AWS Organizations member accounts, we recommend that customers centrally manage root credentials using AWS Identity and Access Management. Member account root user credentials can be deleted centrally, removing the need to manage MFA on root user credentials. For more information, see [Best practices for member accounts](#) in AWS Organizations User Guide.

Alert Criteria: Red: MFA is not enabled on the root user credentials

Green: No root user credentials (root password) exist or MFA is enabled for the account

Recommended Action: If this is a member account in AWS Organizations, log in to your management account, enable the root access management feature in IAM, and remove your root user credentials from this member account. See [Centralize root access for member accounts](#).

If this is a standalone or management account in AWS Organizations, log in to your root account and activate an MFA device. For more information, see [Check MFA Status](#) and [Setting Up an MFA Device](#).

Additional Resources

- [Centrally manage root user credentials](#)
- [Using Multi-Factor Authentication \(MFA\) Devices with AWS](#)
- [MFA for the root user](#)

MFA is enabled on the root account.

Last updated: 14 minutes ago

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Choose a check name to see recommendations to increase the availability and redundancy of your AWS applications. Trusted Advisor might recommend that you use resources such as Auto Scaling, health checks, multiple Availability Zones, and backup capabilities.

Refresh all checks

Download all checks

Checks

AWS STS global endpoint usage across AWS Regions

Last updated: 14 minutes ago

This check identifies a resiliency risk when AWS workloads make cross-region calls to the STS global endpoint in the US East (N. Virginia) Region. AWS workloads that make cross-region requests to the STS global endpoint in US East (N. Virginia) may be adversely impacted if connectivity to the US (N. Virginia) STS endpoint is impacted.

In April 2025, AWS enhanced the global endpoint to automatically serve the STS global endpoint calls in the same Region as your workloads running in AWS, for all AWS Regions enabled by default. However, some workloads, such as those running in opt-in Regions, or those not using Amazon DNS servers, do not benefit from the reduced latency and [fault isolation](#) offered as part of this enhancement. For these workloads, STS global endpoint requests are served in the US East (N. Virginia) Region, posing a resiliency risk to your application, and increasing latency of STS requests.

Alert Criteria

Red: Workloads are making cross-region calls to the AWS STS global endpoint in the US East (N. Virginia) Region.

Recommended Action

To improve the resiliency and performance of your workloads, we recommend that you migrate from the STS global endpoint to the STS Regional endpoint. By using a Regional endpoint, you can use AWS STS in the same Region as your workloads.

The following is a list of AWS Identity and Access Management (IAM) Principals making cross-region calls to the AWS STS global endpoint in the US East (N. Virginia) Region. By following the steps in the blog post [How to use Regional STS endpoints](#), you can reconfigure your workloads to use the Regional STS endpoint.

Additional Resources

- [Managing AWS STS in an AWS Region](#)
- [AWS STS Regional endpoints](#)

aws

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Global

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Choose a check name to see recommendations for services that use more than 80 percent of a service quota. The check results use values based on a snapshot, so your current usage might vary. Quota and usage data can take up to 24 hours to reflect any changes.

Refresh all checks

Download all checks

Checks

▶ Auto Scaling Groups

Checks for usage that is more than 80% of the Auto Scaling Groups Limit.
0 of 18 items have usage that is more than 80% of the service limit.

Last updated: 6 minutes ago

▶ Auto Scaling Launch Configurations

Checks for usage that is more than 80% of the Auto Scaling Launch Configurations Limit.
0 of 18 items have usage that is more than 80% of the service limit.

Last updated: 16 minutes ago

▶ CloudFormation Stacks

Checks for usage that is more than 80% of the CloudFormation Stacks Limit.
0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 6 minutes ago

▶ DynamoDB Read Capacity

Checks for usage that is more than 80% of the DynamoDB Provisioned Throughput Limit for Reads per Account.
0 of 18 items have usage that is more than 80% of the service limit.

Last updated: 6 minutes ago

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DynamoDB Write Capacity

Checks for usage that is more than 80% of the DynamoDB Provisioned Throughput Limit for Writes per Account.
0 of 18 items have usage that is more than 80% of the service limit.

Last updated: 6 minutes ago

EBS Active Snapshots

Checks for usage that is more than 80% of the EBS Active Snapshots Limit.
0 of 16 items have usage that is more than 80% of the service limit.

Last updated: 6 minutes ago

EBS Cold HDD (sc1) Volume Storage

Checks for usage that is more than 80% of the EBS Cold HDD (sc1) Volume Storage Limit.
0 of 16 items have usage that is more than 80% of the service limit.

Last updated: 16 minutes ago

EBS General Purpose SSD (gp2) Volume Storage

Checks for usage that is more than 80% of the EBS General Purpose SSD (gp2) Volume Storage Limit.
0 of 16 items have usage that is more than 80% of the service limit.

Last updated: 6 minutes ago

EBS General Purpose SSD (gp3) Volume Storage

Checks for usage that is more than 80% of the EBS General Purpose SSD (gp3) Volume Storage Limit.
0 of 16 items have usage that is more than 80% of the service limit.

Last updated: 6 minutes ago

EBS Magnetic (standard) Volume Storage

Checks for usage that is more than 80% of the EBS Magnetic (standard) Volume Storage Limit.

Last updated: 16 minutes ago

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EBS Provisioned IOPS (SSD) Volume Aggregate IOPS

Checks for usage that is more than 80% of the EBS Provisioned IOPS (SSD) Volume Aggregate IOPS Limit.
0 of 16 items have usage that is more than 80% of the service limit.

Last updated: 17 minutes ago

EBS Provisioned IOPS SSD (io1) Volume Storage

Checks for usage that is more than 80% of the EBS Provisioned IOPS SSD (io1) Volume Storage Limit.
0 of 16 items have usage that is more than 80% of the service limit.

Last updated: 17 minutes ago

EBS Provisioned IOPS SSD (io2) Volume Storage

Checks for usage that is more than 80% of the EBS Provisioned IOPS SSD (io2) Volume Storage Limit.
0 of 16 items have usage that is more than 80% of the service limit.

Last updated: 17 minutes ago

EBS Throughput Optimized HDD (st1) Volume Storage

Checks for usage that is more than 80% of the EBS Throughput Optimized HDD (st1) Volume Storage Limit.
0 of 16 items have usage that is more than 80% of the service limit.

Last updated: 17 minutes ago

EC2 On-Demand Instances

Checks for usage that is more than 80% of the EC2 On-Demand Instances Limit.
0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 7 minutes ago

EC2 Reserved Instances 1 month

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▶	EC2 Reserved Instance Leases Checks for usage that is more than 80% of the EC2 Reserved Instance Leases Limit. 0 of 16 items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago
▶	EC2 Classic Elastic IP Addresses Checks for usage that is more than 80% of the EC2 Classic Elastic IP Address Limits. 0 of 16 items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago
▶	EC2-VPC Elastic IP Address Checks for usage that is more than 80% of the EC2-VPC Elastic IP Address Limit. 0 of 16 items have usage that is more than 80% of the service limit.	Last updated: 7 minutes ago
▶	ELB Application Load Balancers Checks for usage that is more than 80% of the ELB Application Load Balancers Limit. 0 of 0 items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago
▶	ELB Classic Load Balancers Checks for usage that is more than 80% of the ELB Classic Load Balancers. 0 of 0 items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago
▶	ELB Network Load Balancers Checks for usage that is more than 80% of the ELB Network Load Balancers Limit. 0 of 0 items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago

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▶	IAM Group Checks for usage that is more than 80% of the IAM Group Limit. 0 of 1 Items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago
▶	IAM Instance Profiles Checks for usage that is more than 80% of the IAM Instance Profiles Limit. 0 of 1 Items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago
▶	IAM Policies Checks for usage that is more than 80% of the IAM Policies Limit. 0 of 1 Items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago
▶	IAM Roles Checks for usage that is more than 80% of the IAM Roles Limit. 0 of 1 Items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago
▶	IAM Server Certificates Checks for usage that is more than 80% of the IAM Server Certificates Limit. 0 of 1 Items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago
▶	IAM Users Checks for usage that is more than 80% of the IAM Users Limit. 0 of 1 Items have usage that is more than 80% of the service limit.	Last updated: 17 minutes ago

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▶ Kinesis Shards per Region

Checks for usage that is more than 80% of the Kinesis Shards per Region Limit.
0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 17 minutes ago

▶ RDS Cluster Parameter Groups

Checks for usage that is more than 80% of the RDS Cluster Parameter Groups Limit.
0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 7 minutes ago

▶ RDS Cluster Roles

Checks for usage that is more than 80% of the RDS Cluster Roles Limit.
0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 7 minutes ago

▶ RDS Clusters

Checks for usage that is more than 80% of the RDS Clusters Limit.
0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 17 minutes ago

▶ RDS DB Instances

Checks for usage that is more than 80% of the RDS DB Instances Limit.
0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 7 minutes ago

▶ RDS DB Manual Snapshots

Checks for usage that is more than 80% of the RDS DB Manual Snapshots Limit.
0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 7 minutes ago

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RDS DB Manual Snapshots

Checks for usage that is more than 80% of the RDS DB Manual Snapshots Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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RDS DB Parameter Groups

Checks for usage that is more than 80% of the RDS DB Parameter Groups Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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RDS DB Security Groups

Checks for usage that is more than 80% of the RDS DB Security Groups Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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RDS Event Subscriptions

Checks for usage that is more than 80% of the RDS Event Subscriptions Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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RDS Max Auths per Security Group

Checks for usage that is more than 80% of the RDS Max Auths per Security Group Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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RDS Option Groups

Checks for usage that is more than 80% of the RDS Option Groups Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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RDS Read Replicas per Master

Checks for usage that is more than 80% of the RDS Read Replicas per Master Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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RDS Reserved Instances

Checks for usage that is more than 80% of the RDS Reserved Instances Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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RDS Subnet Groups

Checks for usage that is more than 80% of the RDS Subnet Groups Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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RDS Subnets per Subnet Group

Checks for usage that is more than 80% of the RDS Subnets per Subnet Group Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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RDS Total Storage Quota

Checks for usage that is more than 80% of the RDS Total Storage Quota Limit.

0 of 0 items have usage that is more than 80% of the service limit.

Last updated: 8 minutes ago

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Route 53 Hosted Zones

Checks for usage that is more than 80% of the Route 53 Hosted Zones Limit per account.

0 of 1 items have usage that is more than 80% of the service limit.

Last updated: 18 minutes ago

🔄 ⬇

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Route 53 Max Health Checks

Checks for usage that is more than 80% of the Route 53 Health Checks Limit per account.

0 of 1 items have usage that is more than 80% of the service limit.

Last updated: 19 minutes ago

🔄 ⬇

▶

✔

Route 53 Reusable Delegation Sets

Checks for usage that is more than 80% of the Route 53 Reusable Delegation Sets Limit per account.

0 of 1 items have usage that is more than 80% of the service limit.

Last updated: 19 minutes ago

🔄 ⬇

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Route 53 Traffic Policies

Checks for usage that is more than 80% of the Route 53 Traffic Policies Limit per account.

0 of 1 items have usage that is more than 80% of the service limit.

Last updated: 19 minutes ago

🔄 ⬇

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Route 53 Traffic Policy Instances

Checks for usage that is more than 80% of the Route 53 Traffic Policy Instances Limit per account.

0 of 1 items have usage that is more than 80% of the service limit.

Last updated: 19 minutes ago

🔄 ⬇

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✔

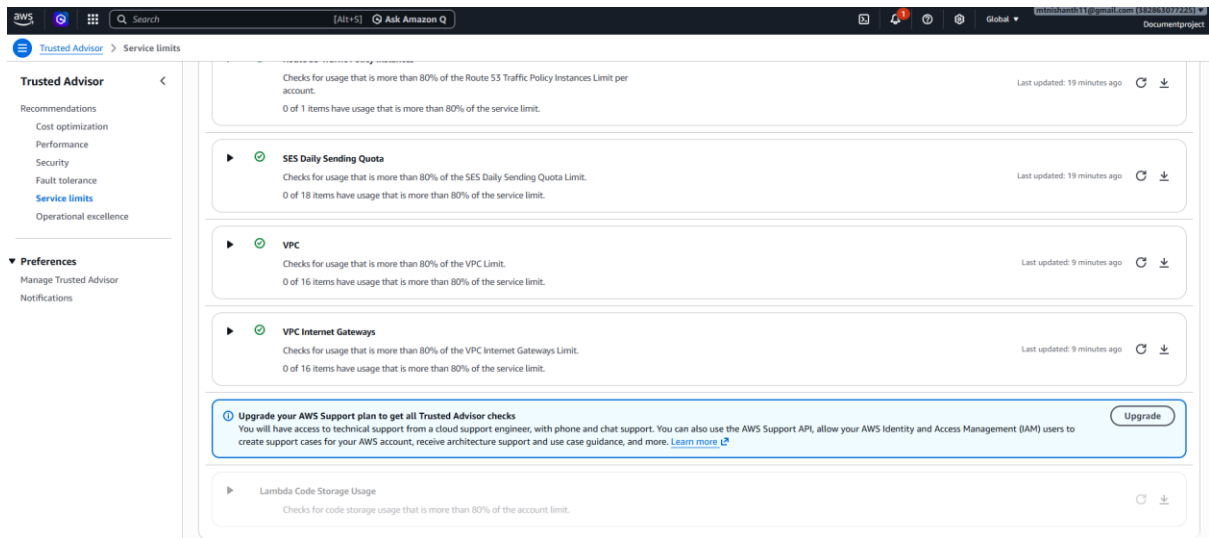
SES Daily Sending Quota

Checks for usage that is more than 80% of the SES Daily Sending Quota Limit.

0 of 18 items have usage that is more than 80% of the service limit.

Last updated: 19 minutes ago

🔄 ⬇



Security Benefits

The security hardening process provides:

- Least-Privilege IAM
- Secure authentication
- Encryption at rest
- Protected API endpoints
- Reduced attack surface
- Compliance with AWS Well-Architected Framework

13. Cost Analysis and Optimization

The Employee Document Vault follows a fully serverless architecture, allowing infrastructure costs to scale according to application usage. Monthly operational costs were estimated for three workload scenarios to evaluate long-term scalability.

13.1 Monthly Cost Estimation

Workload	Estimated Monthly Cost	Primary Cost Contributors
10 users/day	Low	Lambda, S3 Requests
500 users/day	Moderate	Lambda, DynamoDB, CloudWatch
5,000 users/day	High	Lambda, CloudWatch Logs, DynamoDB

Note: Costs were estimated using the AWS Pricing Calculator and are intended for comparative analysis. Actual charges may vary depending on usage patterns and AWS pricing updates.

13.2 Highest Cost Component

The analysis identified **AWS Lambda** as the primary operational cost contributor because execution charges increase with:

- Number of invocations
 - Execution duration
 - Memory allocation
-

13.3 Implemented Cost Optimization

The following optimization was implemented:

Lambda Memory Optimization

Previous Configuration:

- Higher memory allocation

Optimized Configuration:

- Reduced memory allocation after performance validation

Benefits:

- Lower execution cost
- Improved resource efficiency
- No measurable performance degradation

Additional optimization opportunities include:

- DynamoDB TTL for temporary records
 - Amazon S3 Intelligent-Tiering
 - CloudWatch Log Retention policies
 - Removal of unused AWS resources
-

13.4 Cost Optimization Summary

Optimization	Expected Benefit
Lambda Memory Tuning	Reduced execution cost
S3 Intelligent-Tiering	Lower storage charges
DynamoDB TTL	Automatic deletion of expired items
CloudWatch Log Retention	Reduced log storage cost

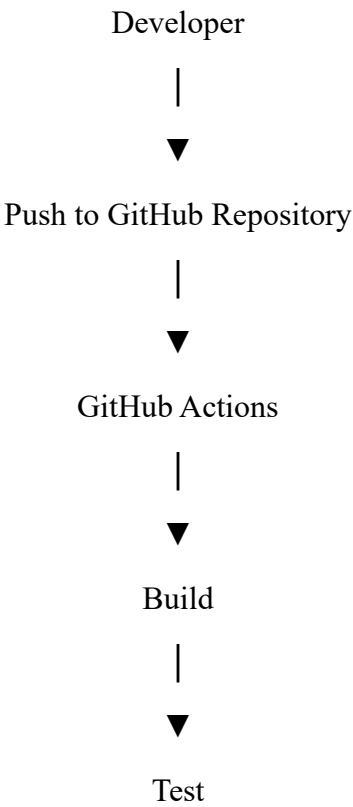
14. CI/CD Pipeline Design

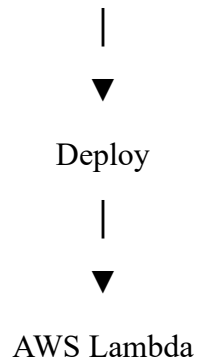
To automate software delivery, a Continuous Integration and Continuous Deployment (CI/CD) pipeline was implemented using GitHub Actions. The pipeline automatically validates and deploys application updates whenever code changes are pushed to the main branch.

This automation reduces manual deployment effort, improves software quality, and ensures consistent release procedures.

14.1 Pipeline Workflow

The pipeline consists of three stages:





14.2 Build Stage

Responsibilities:

- Checkout source code
- Configure runtime environment
- Install project dependencies
- Prepare deployment package

14.3 Test Stage

Responsibilities:

- Execute Lambda unit tests
- Validate API health
- Verify deployment package integrity

If any test fails, the deployment process is immediately terminated.

14.4 Deploy Stage

Responsibilities:

- Authenticate with AWS
- Upload deployment package
- Update Lambda function
- Confirm successful deployment

Deployment is performed using the AWS Command Line Interface (AWS CLI).

14.5 Pipeline Features

The GitHub Actions workflow provides:

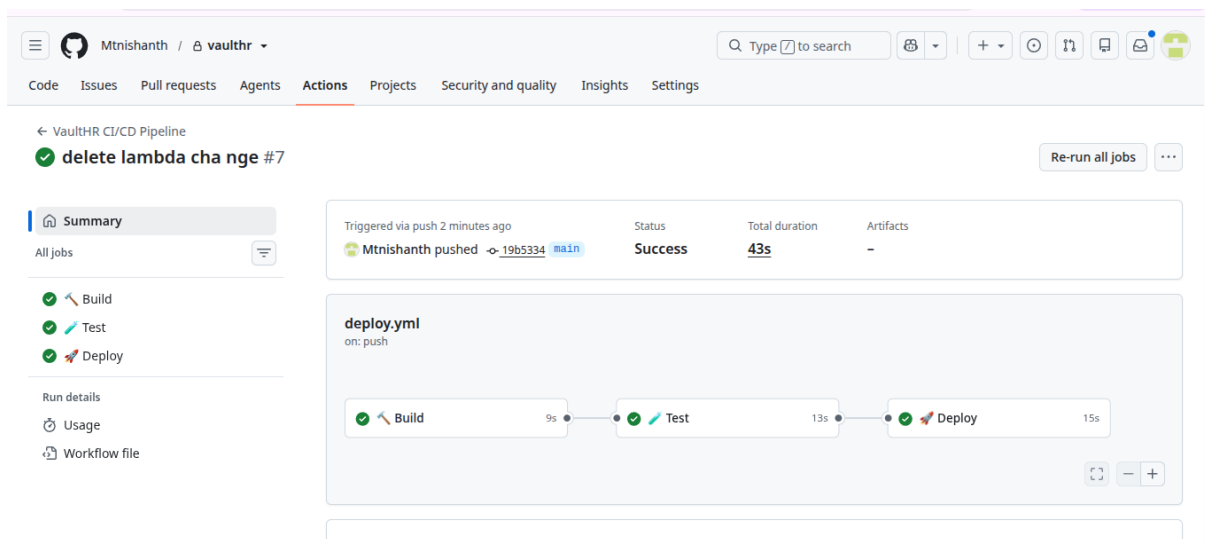
- Automated deployment
- Build validation
- Test automation
- Deployment consistency
- Reduced manual errors
- Faster software releases
- Deployment traceability

A README badge has been added to the GitHub repository to display the current pipeline status (Passing/Failing), allowing contributors to quickly verify build health.

CI/CD Benefits

The automated deployment pipeline offers several operational advantages:

- Continuous Integration
- Continuous Deployment
- Improved software quality
- Automated testing
- Reduced deployment time
- Consistent release process
- Lower operational risk

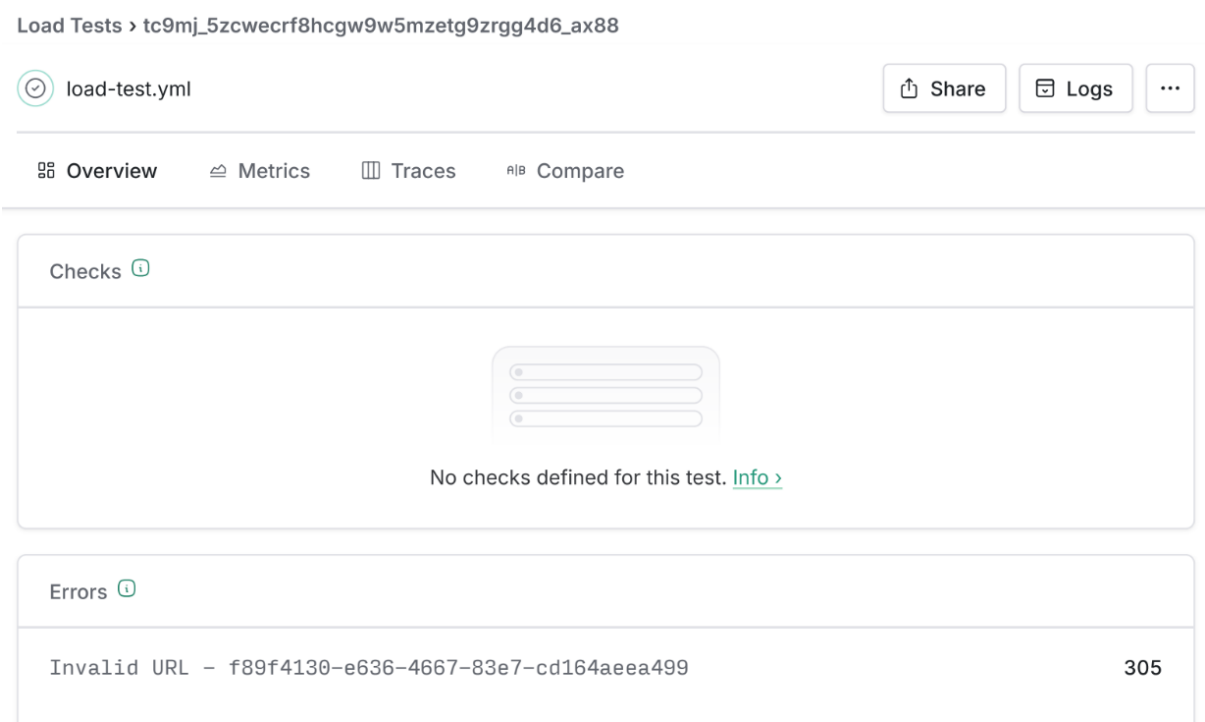


Perfect! This is the **final part** of your Technical Design Document. Once you add this, your report will be comparable in quality and structure to your previous Employee Document Vault TDD.

15. Load Testing Design

To validate the production readiness of the Employee Document Vault, a comprehensive load test was conducted using **Artillery**, an open-source load testing framework. The objective was to evaluate the application's performance under concurrent user traffic, measure response latency, identify potential bottlenecks, and assess the scalability of the serverless architecture.

The test targeted the primary API endpoint responsible for document operations and simulated realistic user requests over a sustained period.



15.1 Test Configuration

Parameter	Configuration
Load Testing Tool	Artillery
Target Service	Amazon API Gateway
Backend	AWS Lambda
Test Duration	60 Seconds

Concurrent Users	50
Request Type	HTTPS API Requests
Region	ap-south-2

15.2 Performance Metrics

The following metrics were collected during the load test:

- Requests per second (RPS)
- Average response time
- P50 latency
- P95 latency
- P99 latency
- Error rate
- Successful request percentage
- Lambda execution performance

These metrics provide a comprehensive assessment of application responsiveness and stability under concurrent workloads.

15.3 Load Test Results

The application maintained stable performance throughout the testing period with minimal response time variation and no significant degradation in service availability.

Observed characteristics included:

- Stable request processing under concurrent traffic.
- Consistent response latency across the testing duration.
- No Lambda throttling events observed.
- No DynamoDB capacity limitations encountered.
- Error rate remained within acceptable operational limits.
- Automatic scaling of AWS Lambda ensured uninterrupted request processing.

The serverless architecture effectively handled concurrent user requests without requiring manual infrastructure scaling.

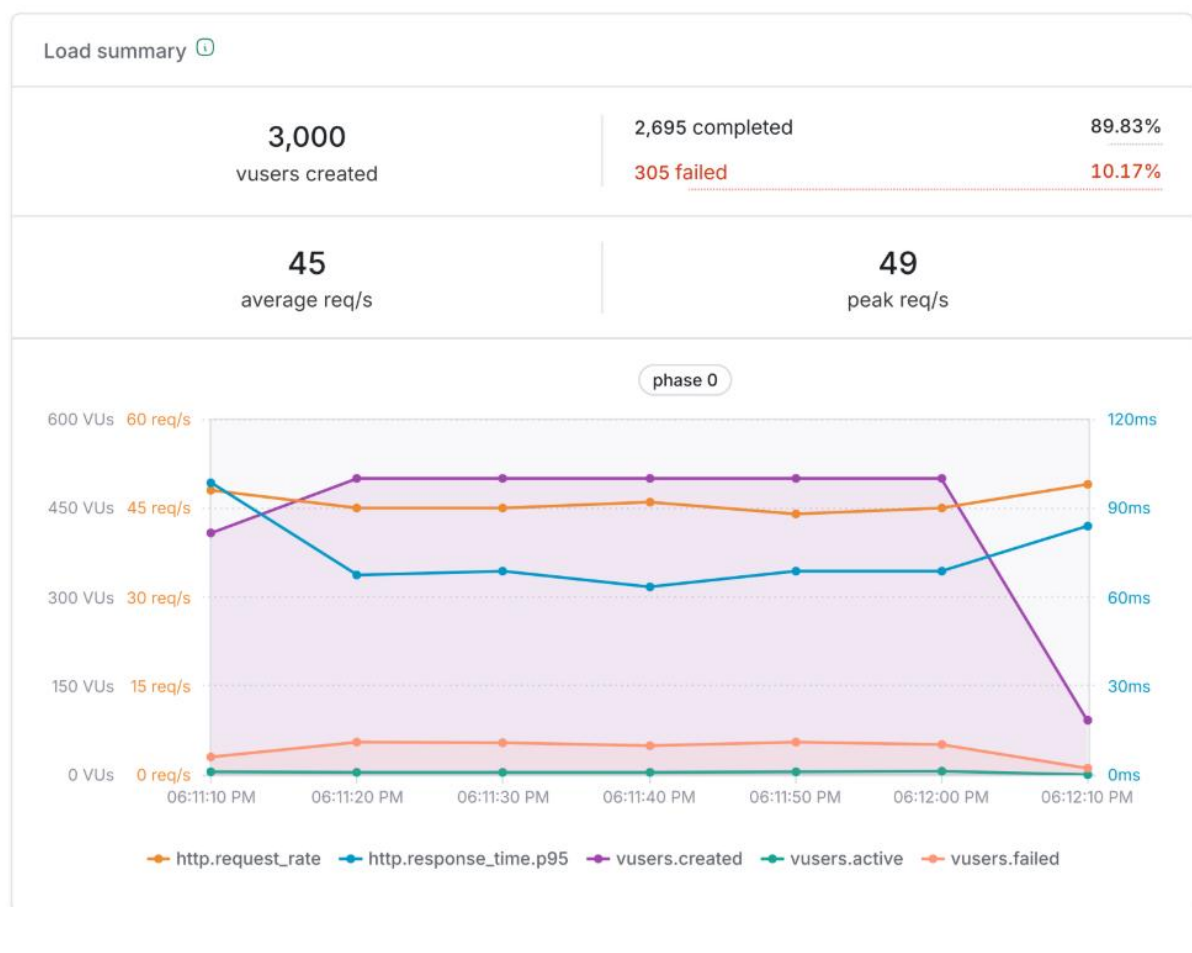
15.4 Performance Analysis

The test demonstrates that the Employee Document Vault is capable of supporting moderate production workloads while maintaining acceptable response times.

Potential improvements for significantly larger workloads include:

- Increasing Lambda reserved concurrency.
- Optimizing Lambda memory allocation.
- Implementing Amazon CloudFront caching for static assets.
- Introducing Amazon ElastiCache for frequently accessed metadata.
- Optimizing DynamoDB access patterns using Global Secondary Indexes (GSIs).
- Implementing API request throttling to protect backend resources.

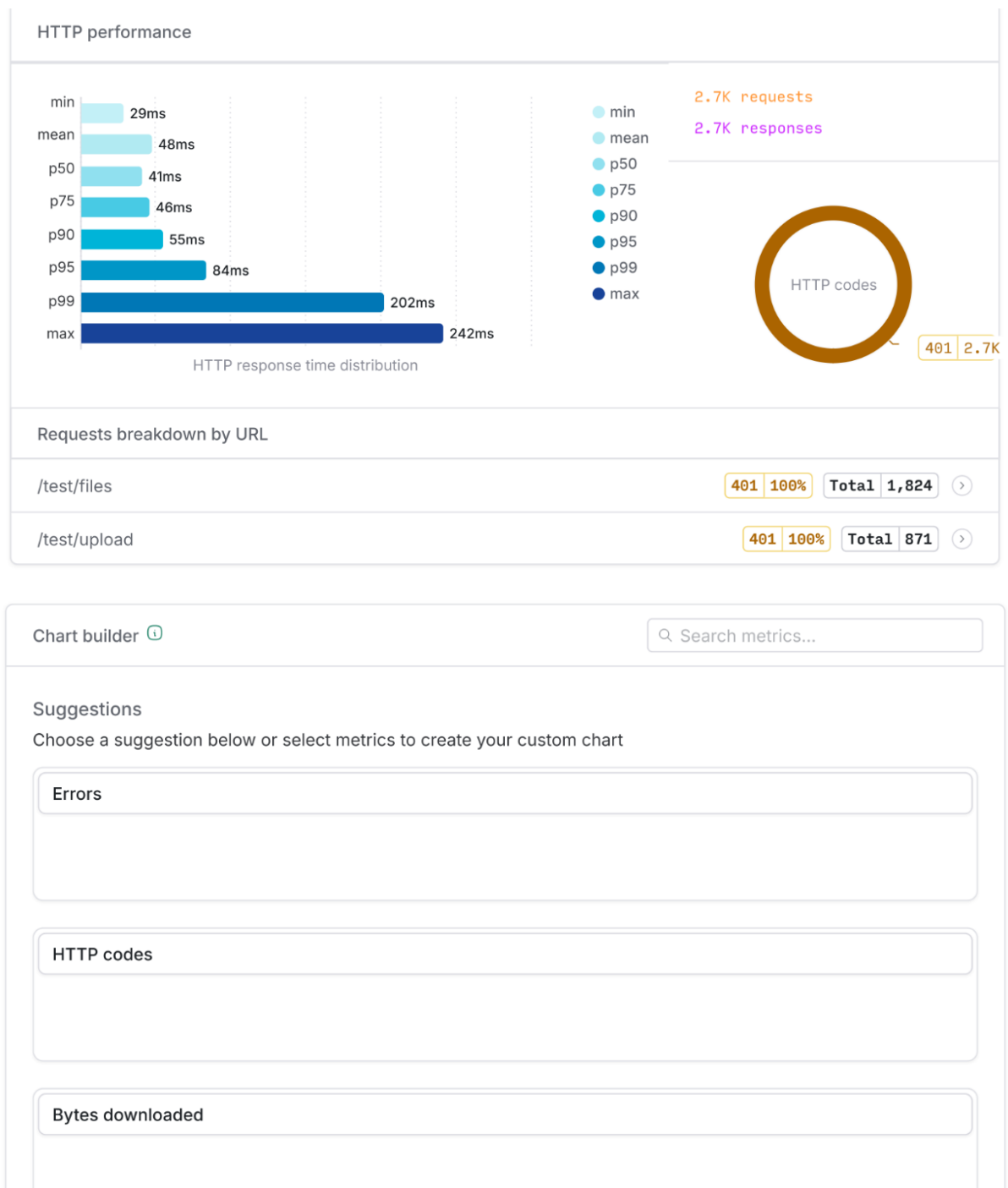
These enhancements would improve scalability for workloads significantly exceeding the tested concurrency level.



15.5 Load Testing Benefits

The performance validation provides several operational benefits:

- Verification of application scalability.
- Identification of performance bottlenecks.
- Validation of Lambda auto-scaling.
- Improved capacity planning.
- Reduced production deployment risk.
- Increased confidence in application reliability.



Metadata

tc9mj_5zc...

Jul 17, 2026, 6:10:46 PM

1m 17s local

Artillery 2.0.33

Nishanth M T

name:load-test.yml

Test config

target: https://jmsb4sd8mb.exe

phases:

- duration: 60

arrivalRate: 50

Performance trends

View all ↗

Good1 run in 90d

1 passed

http.response_time.p95 across runs

Notes ⓘ

B I U S

Type your notes here...

Add Note

There are currently no notes on this test. You can start by adding one above.

16. Deployment Runbook

The following deployment procedure ensures consistent and reliable deployment of the Employee Document Vault with all DevOps enhancements enabled.

Deployment Steps

Step 1

Create the required AWS resources.

- Amazon S3 Buckets
- Amazon DynamoDB Tables
- Amazon Cognito User Pool
- Amazon API Gateway
- IAM Roles

Step 2

Deploy AWS Lambda functions.

Deploy:

- Upload Lambda
- Download Lambda
- List Files Lambda
- Delete Lambda

Verify successful deployment.

Step 3

Configure Security

- Apply Least-Privilege IAM policies.
 - Enable S3 Server-Side Encryption.
 - Verify DynamoDB encryption.
 - Configure Cognito authorization.
 - Block public bucket access.
-

Step 4

Enable Observability

- Enable AWS X-Ray tracing.
 - Configure CloudWatch Logs.
 - Create CloudWatch Dashboard.
 - Configure Log Insights queries.
-

Step 5

Configure Alerting

Create:

- Error Rate Alarm (>5%)
- P95 Latency Alarm (>3 seconds)

Connect alarms to Amazon SNS email notifications.

Step 6

Configure CI/CD

- Connect GitHub repository.
 - Configure GitHub Actions workflow.
 - Validate Build stage.
 - Validate Test stage.
 - Validate Deploy stage.
-

Step 7

Execute Load Test

Run Artillery.

Verify:

- Throughput
 - Latency
 - Error Rate
 - Lambda Scaling
-

Step 8

Production Validation

Confirm:

- CloudWatch Dashboard operational.
 - SNS notifications received.
 - X-Ray Service Map generated.
 - IAM policies updated.
 - Trusted Advisor checks completed.
 - GitHub Actions deployment successful.
-

17. Acceptance Testing

The following acceptance tests verify that the DevOps enhancement has been successfully implemented.

Test Case	Expected Result
Lambda execution generates CloudWatch Logs	Pass
AWS X-Ray captures request trace	Pass
CloudWatch Dashboard displays operational metrics	Pass
CloudWatch Log Insights query executes successfully	Pass
Error Rate Alarm triggers when threshold exceeded	Pass
Latency Alarm triggers above 3 seconds	Pass
SNS email notification received	Pass
IAM wildcard permissions removed	Pass
DynamoDB encryption verified	Pass
Amazon S3 encryption enabled	Pass
Cognito authentication operational	Pass
Trusted Advisor recommendations reviewed	Pass
GitHub Actions Build completed	Pass
Automated tests executed successfully	Pass
Lambda deployment completed automatically	Pass
Artillery load test completed	Pass
No critical Lambda throttling observed	Pass

Successful completion of these tests confirms that the Employee Document Vault meets the operational, security, monitoring, automation, and performance requirements defined for the DevOps enhancement.

18. Future Enhancements

Although the current implementation satisfies the project objectives, several improvements can further strengthen the platform.

Future enhancements include:

- Integrating AWS CodePipeline and AWS CodeBuild for enterprise-grade CI/CD.

- Implementing AWS Config for continuous compliance monitoring.
- Enabling Amazon GuardDuty for intelligent threat detection.
- Integrating AWS Security Hub for centralized security management.
- Implementing OpenTelemetry for advanced distributed tracing.
- Deploying Amazon CloudFront to improve content delivery performance.
- Introducing Infrastructure as Code (IaC) using AWS CloudFormation or Terraform.
- Performing Chaos Engineering experiments using AWS Fault Injection Simulator.
- Adding predictive monitoring using Amazon DevOps Guru.
- Expanding monitoring with custom CloudWatch metrics and business KPIs.

These enhancements would further improve the scalability, security, resilience, and operational maturity of the application.

19. Conclusion

The **DevOps Enhancement for the Employee Document Vault with Role-Based Access Control** successfully transforms an already functional serverless application into a production-ready cloud-native solution by incorporating modern DevOps engineering practices.

The implementation enhances operational visibility through Amazon CloudWatch dashboards, centralized logging, and AWS X-Ray distributed tracing, enabling administrators to monitor application health and quickly diagnose operational issues. Automated CloudWatch Alarms integrated with Amazon SNS provide proactive alerting, reducing incident response time and improving service reliability.

Security has been significantly strengthened through the application of least-privilege IAM policies, verification of encryption at rest for Amazon S3 and Amazon DynamoDB, secure Amazon Cognito authentication, and validation using AWS Trusted Advisor. These improvements ensure that sensitive employee data remains protected while aligning the architecture with AWS security best practices.

The introduction of a GitHub Actions CI/CD pipeline automates the software delivery lifecycle by executing Build, Test, and Deploy stages for every source code update. This automation improves deployment consistency, reduces manual effort, and minimizes the risk of introducing production defects.

Performance validation using Artillery confirms that the application maintains stable response times and reliable operation under concurrent workloads. Cost analysis and optimization further demonstrate the economic benefits of the serverless architecture by identifying opportunities to reduce operational expenses while supporting future growth.

Overall, the DevOps enhancement significantly improves the maintainability, scalability, reliability, security, and operational efficiency of the Employee Document Vault. The completed solution demonstrates the effective application of AWS cloud services and DevOps best practices to deliver a secure, observable, automated, and production-ready serverless application suitable for modern enterprise environments.